Knotted-coil validation report

Generated 2026-10-20T20:10:36.599Z · code git c405dbe · criteria pre-registered in Design.md v1.0 §7 (git 39a4e83). Wire: placeholder d = 1 mm, insulation 0.25 mm.

All pre-registered checks pass: NO

ID	Check	Criterion	Verdict	Summary
V1	Mutual inductance, coaxial circles (r = 50, 40 mm; 400 points) vs Maxwell	relative error < 0.5% at all 5 separations	**PASS**	worst 0.0052%
V2	Self-inductance, circular loop R = 50 mm (400 points) vs mu0 R (ln(8R/a) - 2)	relative error < 2% for R/a >= 50	**PASS**	worst 0.656%
V3	Zero-coupling centre spacing, two coplanar circles	within 1% of the VERIFIED literature value	**NOT EVALUABLE**	computed spacing = 0.7604 x diameter; no VERIFIED reference value available
V4	Convergence of k, trefoil pair (Rf = 40 mm), 300 -> 600 points (final-report resolution)	|k(2n) - k(n)| < 0.001	**PASS**	worst change 1.10e-4
V5	Symmetry: swap of the two elements, and rotation of both by 2pi/q	relative difference < 1e-9 (machine-precision level)	**PASS**	worst 3.13e-14
V6	Topology and clearance across bloom beta = 0.02–3.0, Design v1.2 outward bloom (catalogue knots, core R = 30 mm, placeholder wire)	crossing count and over/under pattern unchanged; minimum self-clearance >= d_w + 2 t_i	**FAIL**	unknot (2-turn) T(2,1) fails at beta=0.02; trefoil T(3,2) fails at beta=0.02,0.05; trefoil T(2,3) fails at beta=0.02,0.05; T(3,4) = 8_19 T(3,4) fails at beta=0.02,0.05; cinquefoil T(2,5) fails at beta=0.02,0.05; T(5,6) T(5,6) fails at beta=0.02,0.05; T(2,7) = 7_1 T(2,7) fails at beta=0.02,0.05; T(3,8) T(3,8) fails at beta=0.02,0.05
V7	Topology and clearance across petal sharpness m = 1–16 x bloom 0.1–3.0 (catalogue knots,	crossing count = q(p-1); pattern equal to m=1, beta=0.4; lift solvable with	**PASS**	all knots pass at all 45 (m, beta) points

	core R = 30 mm, placeholder wire)	clearance >= d_w + 2 t_i		
V8	B1 (Biot–Savart) on the axis of a circular loop, R = 50 mm, 400 points, vs closed form	relative error < 0.5% at z = 0, 10, 20, 40, 80 mm	**PASS**	worst 0.0051%
PH2	POST-HOC: V6 test from the physical bloom floor (outward shape: c / (R sin(pi/p))) upward	post-hoc; not pre-registered (informational)	**AGREES**	T(2,1) from 0.055; T(3,2) from 0.06; T(2,3) from 0.055; T(3,4) from 0.06; T(2,5) from 0.055; T(5,6) from 0.09; T(2,7) from 0.055; T(3,8) from 0.06
PH1	POST-HOC: Maxwell implementation vs Rosa & Grover Example 1 (A = a = 25 cm, d = 20 cm -> 167.08562 cm = nH)	post-hoc; not pre-registered (informational)	**AGREES**	167.08578 nH, rel diff 9.79e-7

V1

Reference: Rosa & Grover 1912, Bull. Bureau of Standards 8:1–237, p. 6 eq. [1] — VERIFIED


```
[
 {
  "d_mm": 5,
  "numeric_nH": 84.42789870378849,
  "maxwell_nH": 84.42735615046593,
  "relErr": 0.000006426273986284069 5
 },
 {
  "d_mm": 10,
  "numeric_nH": 72.20059279228553,
  "maxwell_nH": 72.20064021902708,
  "relErr": -6.56874252107613e-7
 },
 {
  "d_mm": 20,
  "numeric_nH": 49.7568416159741,
  "maxwell_nH": 49.75747180843788,
  "relErr": -0.00001266528303930163 7
 },
 {
  "d_mm": 40,
  "numeric_nH": 24.790747399010364,
  "maxwell_nH": 24.79148544042284,
  "relErr": -0.00002976995526342960 3
 },
 {
  "d_mm": 80,
  "numeric_nH": 7.890639914376564,
  "maxwell_nH": 7.891049880966221,
  "relErr": -0.00005195336436099708
 }
]
```

V2

Reference: Rosa & Grover 1912, p. 111 eq. [62] (surface current; leading term) — VERIFIED


```
[
 {
  "R_over_a": 50,
  "numeric_nH": 250.24750941864468,
  "closedForm_nH": 250.79111396517092,
  "relErr": -0.002167559041193675
 },
 {
  "R_over_a": 100,
  "numeric_nH": 293.24137207410735,
  "closedForm_nH": 294.34283577124296,
  "relErr": -0.003742111453976914
 },
 {
  "R_over_a": 200,
  "numeric_nH": 335.6774399998626,
  "closedForm_nH": 337.894557577315,
  "relErr": -0.006561566405061494
 }
]
```


V3

Reference: Roemer et al. 1990 MRM 16:192 — UNVERIFIED (paywalled, not read). Closest primary text read: Roemer & Edelstein, US 4,825,162 (1989), "D approximately equal to 0.75d" (PARAPHRASE-ONLY; Google Patents OCR text). Exploratory comparison only: 1.39% from the patent's approximate value.


```
[
 {
  "computed_spacing_over_diameter": 0.7604234630725386,
  "patent_value": 0.75,
  "diff_vs_patent": 0.013897950763384811
 }
]
```


V4

k160 (stage-2 search resolution) shown for information; the criterion applies to the reported resolution (300).


```
[
 {
  "dOverRf": 1.14,
  "rot": [
   0,
   0
  ],
  "k160": 0.06459163222238361,
  "k300": 0.06455516418643788,
  "k600": 0.06449108409945338,
  "change300to600": 0.00006408008698449708
```

```
 },
 {
  "dOverRf": 1.6,
  "rot": [
   0,
   0
  ],
  "k160": -0.07378139423091604,
  "k300": -0.07357698017927117,
  "k600": -0.07349667406540387,
  "change300to600": 0.0000803061138672928
 },
 {
  "dOverRf": 1.14,
  "rot": [
   0.3,
   0.7
  ],
  "k160": 0.0655351613003449,
  "k300": 0.06549676042575976,
  "k600": 0.06543397784935304,
  "change300to600": 0.00006278257640672524
 },
 {
  "dOverRf": 1.6,
  "rot": [
   0.3,
   0.7
  ],
  "k160": -0.06840015446728545,
  "k300": -0.06810298788665974,
  "k600": -0.06799308943677852,
  "change300to600": 0.00010989844988121278
 }
]
```


V5


```
[
 {
  "knot": "T(2,3)",
  "swapRelDiff": 7.100113546090396e-15,
  "rotRelDiff": 1.1178902178950837e-14
 },
 {
  "knot": "T(3,4)",
  "swapRelDiff": 1.0138156947293846e-14,
  "rotRelDiff": 2.7035085192783586e-15
 },
 {
  "knot": "T(2,5)",
  "swapRelDiff": 4.452427967451506e-15,
  "rotRelDiff": 3.132601105671238e-14
 },
 {
  "knot": "T(3,8)",
  "swapRelDiff": 1.7625397784369575e-14,
  "rotRelDiff": 1.6697745269402758e-14
 }
]
```

V6


```
[
 {
  "knot": "unknot (2-turn) T(2,1)",
  "pass": false,
  "fails": [
   {
    "beta": 0.02,
    "crossings": 1,
    "expected": 1,
    "patternSame": true,
    "liftFeasible": false,
    "clearance": 0.6000000000000014
   }
  ]
 },
 {
  "knot": "trefoil T(3,2)",
  "pass": false,
  "fails": [
   {
    "beta": 0.02,
    "crossings": 4,
    "expected": 4,
    "patternSame": true,
    "liftFeasible": false,
    "clearance": 0.5196152422706664
   },
   {
    "beta": 0.05,
    "crossings": 4,
    "expected": 4,
    "patternSame": true,
    "liftFeasible": false,
    "clearance": 1.2990381056766571
   }
  ]
 },
 {
  "knot": "trefoil T(2,3)",
  "pass": false,
  "fails": [
   {
    "beta": 0.02,
    "crossings": 3,
    "expected": 3,
    "patternSame": true,
    "liftFeasible": false,
    "clearance": 0.5999999999999999
   },
   {
    "beta": 0.05,
    "crossings": 3,
    "expected": 3,
    "patternSame": true,
    "liftFeasible": false,
    "clearance": 1.4999999999999987
   }
  ]
 },
 {
  "knot": "T(3,4) = 8_19 T(3,4)",
  "pass": false,
```

```
  "fails": [
   {
    "beta": 0.02,
    "crossings": 8,
    "expected": 8,
    "patternSame": true,
    "liftFeasible": false,
    "clearance": 0.5196152422706636
   },
   {
    "beta": 0.05,
    "crossings": 8,
    "expected": 8,
    "patternSame": true,
    "liftFeasible": false,
    "clearance": 1.2990381056766551
   }
  ]
 },
 {
  "knot": "cinquefoil T(2,5)",
  "pass": false,
  "fails": [
   {
    "beta": 0.02,
    "crossings": 5,
    "expected": 5,
    "patternSame": true,
    "liftFeasible": false,
    "clearance": 0.6000000000000001
   },
   {
    "beta": 0.05,
    "crossings": 5,
    "expected": 5,
    "patternSame": true,
    "liftFeasible": false,
    "clearance": 1.4999999999999984
   }
  ]
 },
 {
  "knot": "T(5,6) T(5,6)",
  "pass": false,
  "fails": [
   {
    "beta": 0.02,
    "crossings": 24,
    "expected": 24,
    "patternSame": true,
    "liftFeasible": false,
    "clearance": 0.35267115137547855
   },
   {
    "beta": 0.05,
    "crossings": 24,
    "expected": 24,
    "patternSame": true,
    "liftFeasible": false,
    "clearance": 0.8816778784387074
   }
  ]
 },
 {
  "knot": "T(2,7) = 7_1 T(2,7)",
```

```
  "pass": false,
  "fails": [
   {
    "beta": 0.02,
    "crossings": 7,
    "expected": 7,
    "patternSame": true,
    "liftFeasible": false,
    "clearance": 0.5999999999999991
   },
   {
    "beta": 0.05,
    "crossings": 7,
    "expected": 7,
    "patternSame": true,
    "liftFeasible": false,
    "clearance": 1.4999999999999984
   }
  ]
 },
 {
  "knot": "T(3,8) T(3,8)",
  "pass": false,
  "fails": [
   {
    "beta": 0.02,
    "crossings": 16,
    "expected": 16,
    "patternSame": true,
    "liftFeasible": false,
    "clearance": 0.5196152422706642
   },
   {
    "beta": 0.05,
    "crossings": 16,
    "expected": 16,
    "patternSame": true,
    "liftFeasible": false,
    "clearance": 1.299038105676654
   }
  ]
 }
]
```


V7

Tightest bend radius per knot is shown for information only (not part of the criterion).


```
[
 {
  "knot": "unknot (2-turn) T(2,1)",
  "pass": true,
  "fails": [],
  "minBendRadius_info": 30.001517230113095
 },
 {
  "knot": "trefoil T(3,2)",
  "pass": true,
  "fails": [],
  "minBendRadius_info": 30.00320420514439
 },
 {
```

```
  "knot": "trefoil T(2,3)",
  "pass": true,
  "fails": [],
  "minBendRadius_info": 12.679237365454245
 },
 {
  "knot": "T(3,4) = 8_19 T(3,4)",
  "pass": true,
  "fails": [],
  "minBendRadius_info": 18.044503562876002
 },
 {
  "knot": "cinquefoil T(2,5)",
  "pass": true,
  "fails": [],
  "minBendRadius_info": 3.611754690870217
 },
 {
  "knot": "T(5,6) T(5,6)",
  "pass": true,
  "fails": [],
  "minBendRadius_info": 20.009106697467402
 },
 {
  "knot": "T(2,7) = 7_1 T(2,7)",
  "pass": true,
  "fails": [],
  "minBendRadius_info": 1.7530670446917516
 },
 {
  "knot": "T(3,8) T(3,8)",
  "pass": true,
  "fails": [],
  "minBendRadius_info": 3.1285172847350067
 }
]
```


V8

Reference: Closed form DERIVED from Biot–Savart (core/field.js header); not a literature value


```
[
 {
  "z_mm": 0,
  "numeric_uT_per_A": 12.567016602763701,
  "closedForm_uT_per_A": 12.566370614359172,
  "relErr": 0.00005140612388037506
 },
 {
  "z_mm": 10,
  "numeric_uT_per_A": 11.848970938155405,
  "closedForm_uT_per_A": 11.848404026933501,
  "relErr": 0.00004784705354540025
 },
 {
  "z_mm": 20,
  "numeric_uT_per_A": 10.058649199017946,
  "closedForm_uT_per_A": 10.058260521084454,
  "relErr": 0.00003864265920303634
 },
 {
  "z_mm": 40,
```

```
  "numeric_uT_per_A": 5.983437165365943,
  "closedForm_uT_per_A": 5.983345646740566,
  "relErr": 0.00001529556050744994
 },
 {
  "z_mm": 80,
  "numeric_uT_per_A": 1.8708040216387938,
  "closedForm_uT_per_A": 1.870832332948728,
  "relErr": -0.00001513300226619956
 }
]
```


PH2


```
[
 {
  "knot": "T(2,1)",
  "betaMinPhysical": 0.05,
  "testedFrom": 0.055,
  "failsAt": []
 },
 {
  "knot": "T(3,2)",
  "betaMinPhysical": 0.05773502691896258,
  "testedFrom": 0.06,
  "failsAt": []
 },
 {
  "knot": "T(2,3)",
  "betaMinPhysical": 0.05,
  "testedFrom": 0.055,
  "failsAt": []
 },
 {
  "knot": "T(3,4)",
  "betaMinPhysical": 0.05773502691896258,
  "testedFrom": 0.06,
  "failsAt": []
 },
 {
  "knot": "T(2,5)",
  "betaMinPhysical": 0.05,
  "testedFrom": 0.055,
  "failsAt": []
 },
 {
  "knot": "T(5,6)",
  "betaMinPhysical": 0.08506508083520399,
  "testedFrom": 0.09,
  "failsAt": []
 },
 {
  "knot": "T(2,7)",
  "betaMinPhysical": 0.05,
  "testedFrom": 0.055,
  "failsAt": []
 },
 {
  "knot": "T(3,8)",
  "betaMinPhysical": 0.05773502691896258,
  "testedFrom": 0.06,
  "failsAt": []
```

```
  }
]
```

PH1